BIOCHEMISTRY CUME
OCTOBER 29, 2004

Enzymes

Each question weighted equally.

1. Initial velocities have been measured for the reaction of alpha chymotrypsin with tyrosine benzyl ester [S] at six different substrate concentrations. Use the data below to make a reasonable estimate of V_{max} and K_m value for this substrate. (25 %)

mM [S]	0.01	0.04	0.1	2.0	10
v_o (mM/min)	35	56	66	69	70

2. a) Why is the k_{cat}/K_m value (specificity constant) used to measure the preference of an enzyme for different substrates?
   b) What are the upper limits for k_{cat}/K_m values for enzymes?
   c) Enzymes with k_{cat}/K_m values approaching these upper limits are said to have reached "catalytic perfection". Explain. (25%)

3. An Enzyme that follows Michaelis-Menten kinectics has a K_m of 1 μM. The initial velocity is 0.1 μM/min at the substrate concentration of 100 μM. What is the initial velocity when [S] is equal to a) 1 mM, b) 1 μM, or c) 2μ M? (25 %)

4. Draw a graph of v_o versus [S] for a typical enzyme reaction a) in the absence of inhibitor, b) in the presence of competitive inhibitor, and c) in the presence of non-competitive inhibitor. (25 %)